Universidad Nacional de Loja

Faculty of Energy, Industry,
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Electrical Engineering Program

Characterization of Electrical Substations

Transmission Lines and Electrical Substations

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1. Substation Fundamentals

An electrical substation is a key node in a power system. Through switching, control, protection, and transformation equipment, it converts the high-voltage energy received from transmission or subtransmission networks to voltage levels suitable for distribution (CELEC EP TRANSELECTRIC, 2016; Sivanagaraju, 2010).

1.1 Primary Components

The configuration of an electrical substation depends on its type, function, and voltage level. In general, its components are grouped into switchyard equipment, auxiliary services, protection and control systems, and structural elements (CELEC EP TRANSELECTRIC, 2016; Mier Luna, 1983; Naranjo Pilalo, 2022; Sivanagaraju, 2010). The following catalog summarizes the function of the principal components.

Busbars

Main conductors to which incoming and outgoing circuits operating at the same voltage level are connected. They are generally made of copper or aluminum, formed as rigid tubes or flexible conductors, and mounted on insulators.

Insulators

Components that provide mechanical support and electrical isolation for energized conductors and busbars. They are commonly manufactured from porcelain or composite materials.

Disconnect Switches

Mechanical switching devices used to connect or isolate parts of an electrical system under no-load conditions. They provide a visible isolation point for safe maintenance and may be manually or motor operated.

Circuit Breakers

Devices capable of opening and closing circuits under load and interrupting fault currents. They incorporate arc-extinction systems and electrical and mechanical protection mechanisms. Typical interrupting media include vacuum, oil, and sulfur hexafluoride (SF_6).

Reclosers

Protection and switching devices that automatically restore service after temporary faults. They are widely used in distribution networks and operate as circuit breakers with a programmed reclosing sequence.

Power Transformers

Equipment that changes voltage levels without changing system frequency. Depending on the transmission or distribution application, units may be step-up or step-down, single-phase or three-phase, and dry-type or liquid-immersed.

Instrument Transformers

Current transformers (CTs) and voltage transformers (VTs) reduce primary current and voltage to standardized values suitable for meters and protective relays. Their secondary signals enable the detection of overloads, short circuits, and other abnormal conditions so that the affected equipment can be isolated.

Surge Arresters

Protect equipment against lightning impulses, switching surges, and fast-front overvoltages. Arresters must withstand specified temporary overvoltages without operating and safely divert surge current to ground when their protective level is exceeded.

Grounding Grid

A low-impedance network that conducts fault and lightning currents into the earth while controlling touch and step voltages. It must provide adequate conductivity, mechanical integrity, and corrosion resistance throughout the installation's service life.

Auxiliary Services

AC and DC supply systems that power control, protection, signaling, lighting, ventilation, heating, battery-charging, compressor, and fire-protection equipment.

Supervisory and Control System

An integrated system of control panels, protective relays, data-acquisition devices, telecommunications equipment, alarms, and indicators, generally located in the control building, that supports local and remote monitoring and operation.

Medium-Voltage Switchgear

Metal-enclosed assemblies containing protection, metering, and switching equipment for medium-voltage substations. Their compartmentalized construction supports safe, organized power distribution and maintenance.

2. Switching and Isolation Equipment

Standardized identification of switching and isolation devices is essential for rapid, efficient, and safe equipment recognition during substation operation and maintenance. In Ecuador's National Transmission System (SNT), each device is identified by a five-character alphanumeric code that specifies the equipment type, operating voltage, bay location, and function within the substation (Brantes Meza, 2008; Yanza Zamora, 2021).

The code is organized into five positions:

- **Positions 1 and 2—equipment type (ANSI device number):**
  - 52: AC circuit breaker.
  - 89: Disconnect switch.
- **Position 3—operating voltage level:**
  - 0: 69 kV.
  - 1: 138 kV.
  - 2: 230 kV.
  - 5: 500 kV.
  - 7: Capacitor bank.
- **Position 4—equipment bay or position:**
  - 1–9: Lines 1 through n.
  - K, Q, R, U, T: Transformers or autotransformers.
  - W, X: Capacitor banks.
  - ϕ: Bus-coupler or transfer bay.
- **Position 5—equipment function within the bay:**
  - 1: Bay disconnect switch on the bus side.
  - 2: Circuit breaker (applicable only to device code 52).
  - 3: Bay disconnect switch on the line or equipment side.
  - 4: Line-side or post-bus grounding switch.

- 5: Transfer or bypass disconnect switch.
- 6: Bus 1 grounding switch.
- 7: Bus 1 selector disconnect switch.
- 8: Bus 2 grounding switch.
- 9: Bus 2 selector disconnect switch.

3. Disconnect Switches

Disconnect switches are mechanical devices whose primary function is to establish or interrupt circuit continuity when no significant load current is present. They provide a visible isolation gap so that system sections can be safely separated for maintenance or operational switching. Because they do not have arc-extinction chambers, they must normally be operated under no-load conditions (Electro-Genesis, n.d.; Osorio Patiño & Culma Ramírez, 2017).

3.1 General Functions

- Provide visible and reliable circuit isolation.
- Isolate equipment or lines for maintenance.
- Carry normal current and specified abnormal current for a limited duration.
- Coordinate with circuit breakers through interlocking that prevents load-current interruption by the disconnect switch.
- In designated applications, interrupt low-level induced or capacitive currents.

3.2 Main Components

- **Insulating column:** Supports energized parts and isolates them from ground.
- **Blades:** Moving contacts that establish or interrupt continuity with the fixed contacts.
- **Terminals:** Connect the incoming and outgoing conductors.
- **Metal base:** Provides structural support for the assembly.
- **Operating mechanism:** Provides manual or motor-driven actuation, generally through a rotating linkage.

3.3 Classification by Operating Mechanism

3.3.1 Center-Break Disconnect Switches

These switches have two rotating insulator columns with attached blades that rotate in opposite directions, producing a single isolation gap at the center. They are compact and economical, but are less suitable for extra-high-voltage (EHV) applications because the open blades remain energized and cantilevered. This condition requires greater safety clearances and more frequent maintenance. They are commonly applied at medium and high voltages, subject to design limitations at higher voltage levels.

Figure 1. Center-break disconnect switch.

3.3.2 Double-Side-Break Disconnect Switches

These switches use three insulator columns per pole. The moving blades are mounted on the rotating center column, while the fixed contacts are mounted on the outer columns. The arrangement provides two isolation gaps and requires less horizontal space than some alternative designs, making it suitable for space-constrained substations. Although the design is more expensive and its blades may be susceptible to deformation, it provides clear visual isolation and reliable operation. It is used mainly at voltages up to 245 kV.

Figure 2. Double-side-break disconnect switch.

3.3.3 Vertical-Break Disconnect Switches

These switches have three insulator columns per pole. The blade first rotates about its axis and then rises vertically. They are common in EHV substations because they require a narrower bay, although their installation height is greater. Arcing contacts may be included to divert arcs away from the main contacts. They are typically used as bypass switches mounted on gantries.

Figure 3. Vertical-break disconnect switch.

3.3.4 Pantograph Disconnect Switches

Built on a single support insulator, pantograph switches connect busbars installed at different elevations. The pantograph-shaped moving contact rises to engage a suspended fixed contact. Their principal advantage is the small installation footprint; however, installation requires precise mechanical alignment and regular maintenance. The moving contact trajectory must coincide accurately with the fixed contact to ensure proper engagement.

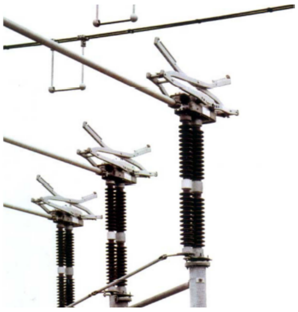

Figure 4. Pantograph disconnect switch.

3.3.5 Semi-Pantograph Disconnect Switches

These switches are similar to vertical-break designs but use a horizontally articulated pantograph arm. They combine a compact footprint with reliable operation and are well suited to EHV substations with limited phase-to-phase spacing. The moving contact is guided into the fixed contact and secured by mechanical guides. They are generally the most expensive option, but require the least structural space.

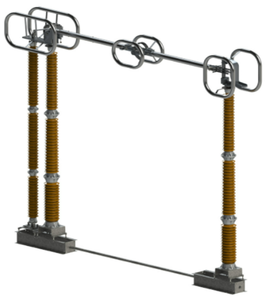

Figure 5. Semi-pantograph disconnect switch.

4. Power Circuit Breakers

A circuit breaker is a switching device designed to open or close an electrical circuit under load, either automatically in response to a fault or through an operator command. When the contacts separate, an electric arc forms and must be extinguished efficiently to prevent equipment damage. The arc-interruption method defines the breaker type; common interrupting media include oil, air, vacuum, and specialized gases (Mier Luna, 1983).

The following sections classify circuit breakers according to their arc-interruption method.

4.1 Oil Circuit Breakers

In bulk-oil circuit breakers, the arc vaporizes the oil and produces gases that cool and deionize the arc path. The design may use a single break or multiple breaks, with multiple-break arrangements applied to breakers with high interrupting ratings. Arc extinction is enhanced by oil circulation caused by pressure differences, although cooling was limited in early designs.

Minimum-oil circuit breakers use an interrupting chamber whose action adapts to the current magnitude. Before the current reaches its natural zero, fresh oil is forced between the contacts, improving interruption even with short contact gaps.

Mixed-quenching circuit breakers combine liquid and solid interrupting materials. Decomposition of the extinguishing material generates a hydrogen-rich gas that cools the arc intensely and promotes deionization.

4.2 Compressed-Air Circuit Breakers

In this type of breaker, an axial jet of high-pressure air traveling at approximately sonic velocity cools and interrupts the arc. Dielectric strength is restored by injecting clean air that removes ionized gases. The compressed air must be dehumidified and cleaned of lubricant residues before entering the interrupting system.

Arreola (2010) notes that opening a pneumatic valve produces a turbulent flow that carries the arc and hot gases into the atmosphere. These breakers can be applied across a broad range of voltage and interrupting ratings, and their quenching effectiveness is largely independent of the interrupted current.

Martín (1987) explains that compressed-air breakers were developed to eliminate the fire risk associated with oil. Typical operating pressures range from 8 to 13 $\mathrm{kg\,cm^{-2}}$, with interruption times on the order of three cycles. Because they can generate switching overvoltages, damping resistors and capacitors are commonly included.

4.3 Atmospheric-Air Magnetic-Blast Circuit Breakers

The arc is divided into a series of shorter arcs between copper plates arranged around an axis. A magnetic field drives the arcs rapidly across the plates, improving cooling and reducing electron emission. After current zero, the cool metal plates help the gap recover its dielectric strength quickly.

4.4 Vacuum Circuit Breakers

Vacuum circuit breakers operate within sealed interrupters maintained at very low pressure. Arc extinction occurs primarily through rapid diffusion of the metal vapor plasma. Because there is no bulk gas to sustain ionization, the arc is extinguished quickly at current zero, usually within one cycle.

Martín (1987) notes that the moving contact is actuated through a metallic bellows. These breakers offer fast operation and low maintenance requirements, but they can produce current-chopping overvoltages and will lose interrupting capability if vacuum integrity is compromised.

4.5 SF_6 Circuit Breakers

Sulfur hexafluoride has high dielectric strength, good thermal conductivity, and strong electron-capture capability, properties that support rapid arc extinction and dielectric recovery.

Mier Luna (1983) and Arreola (2010) distinguish among the following designs:

- **Single-pressure (puffer) breakers:** Operate at approximately 3–7 bar and use contact motion to produce a temporary pressure rise in the interrupting chamber.
- **Two-pressure breakers:** Use a closed system in which the gas is compressed and recirculated. Three poles may share a common tank and compressor.
- **Dynamic self-compression interrupters:** Retract the fixed contact near the end of the opening stroke to improve gas-flow control and interruption efficiency.

SF_6 circuit breakers may also be classified by tank arrangement:

- **Dead-tank design:** The interrupting chamber is enclosed in a grounded tank and connected through bushings.
- **Live-tank design:** The interrupting chamber is supported on insulators and remains at line potential.

Martín (1987) reports that SF_6 enables interruption times as short as two cycles and that breakers may use independent-pole or three-pole arrangements. Grading or pre-insertion resistors help control overvoltages associated with rapid interruption.

5. Lightning Counters and Line Traps

5.1 Lightning Counters

A lightning counter is a device designed to record automatically the number of lightning events occurring within a monitored area by detecting rapid electromagnetic or electrostatic field variations produced by the discharges (Villacís Salazar, 1980). Its principal purpose is to monitor lightning activity, particularly ground strikes, and support evaluation of the area's ground-flash density.

Modern devices may include time-stamped event logging, remote monitoring over Ethernet networks and TCP-based protocols, and compatibility with both new and existing lightning-protection systems (García Castillo, 2019). Some models are designed to withstand currents of up to 100 kA and can detect fields over an area of up to 200 m^2 without a direct connection to the grounding system.

The recorded data support assessment of the lightning-protection system and the development of local lightning-density maps used in the planning and maintenance of substations and other sensitive networks.

Figure 6. Lightning counter.

5.2 Line Traps

Line traps, also called high-frequency blocking coils, are essential components of power-line carrier communication systems associated with transmission lines (Sosa Espinosa, 1975). Their primary function is to prevent high-frequency carrier signals from flowing into busbars or other high-voltage equipment. They are connected in series with the power conductor at each carrier-coupling point.

Line traps are installed at both ends of the protected line section, confining the carrier signal to that section. This arrangement increases the useful signal level, reduces interference from adjacent channels and external faults, and improves communication reliability.

Structurally, a line trap consists of a main coil rated for the line's continuous and short-circuit currents. One or more tuning circuits are connected in parallel with the coil to establish the required blocking-frequency response. The resulting network presents high impedance at the selected carrier frequency while maintaining negligible impedance at the power frequency of 50 or 60 Hz, allowing normal power transfer.

The main coil may be cylindrical or constructed in layers and is commonly made from materials compatible with high-voltage conductors, such as aluminum or steel. The tuning unit is installed within or adjacent to the coil and sets the blocking band. A surge arrester connected across the trap protects it against overvoltages.

For discrete carrier frequencies, a line trap may use single- or double-frequency tuning to block one or two carrier bands. Two double-frequency traps may be connected in series when four bands must be blocked. Because series connection of more than three units becomes complex, a wideband line trap is generally preferred; a typical design with an inductance of approximately 2.0 mH can block carrier frequencies from 50 to 300 kHz.

In summary, line traps are passive selective filters that provide high-frequency isolation without interfering with power-frequency operation. Their design depends on the application, the number of frequencies or bands to be blocked, and the electrical characteristics of the transmission line.

Figure 7. Line trap.

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